Nepal Hydropower Data Analysis: EDA and Process Documentation

Project documentation for the notebook `Nepal_Hydropower_Data_Analysis.ipynb`. It describes the dataset, the cleaning and transformation steps, the exploratory data analysis (EDA) results and the key findings.

1. Project Overview

Objective: clean, transform and explore Nepal's hydropower licence register, then turn the results into insights that can guide decisions.

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn and scikit-learn.

Pipeline: data loading and exploration, cleaning and transformation, EDA, feature selection, insights and recommendations.

2. Dataset Overview

The dataset (`hydropower.csv`) has **572 rows and 14 columns**. Each row is one hydropower licence record.

Column	Type	Description
Project	text	Project name
Province, District, Municipality, Address	text	Administrative location
Capacity (MW)	number	Installed or proposed capacity
River	text	River used by the project
Lic No	text	Licence number
Isuue Date, Validity	text	Issue and expiry dates (Bikram Sambat calendar)
Promoter	text	Developer or owner
Longitude, Latitude	number	Project coordinates

Column	Type	Description
License Type	text	Survey, Generation or Operation

There are 3 numeric and 11 text columns. The data covers 6 provinces, 54 districts, 186 municipalities, 315 rivers and 511 distinct promoters.

3. Initial Exploration and Data-Quality Findings

The raw file contains no `NaN` values and no exact duplicate rows. Coordinates and capacities are within valid ranges, with no capacity at or below zero. However, a closer inspection revealed hidden problems.

Issue	Evidence	Treatment
Placeholder licence numbers	`Lic No` is `0` in 48 rows and `1_gtd` in 12 rows; these cause the 60 repeated licence numbers	Converted to missing, labelled `Unknown`
Placeholder addresses	`Address` is `0` in 42 rows	Converted to missing, labelled `Unknown`
Placeholder validity	`Validity` is `0/1/0` in 40 rows	Converted to missing
Dates stored as text	Date columns are `object` type	Parsed into numeric parts
Bikram Sambat calendar	Years such as 2073	Kept as BS year, approximate AD year derived
Extreme skew	Mean 41.9 MW, median 9.2 MW, maximum 1,902 MW	Outliers flagged, log transform added
Repeated project names	3 names appear twice	Inspected and kept
Province 2 absent	Only provinces 1, 3, 4, 5, 6 and 7 appear	Documented as a limitation

The column header `Isuue Date` is a typo in the source and was renamed `Issue Date`.

4. Data Cleaning and Transformation

4.1 Standardization

Whitespace was stripped from all text columns, district names were upper-cased, licence types were title-cased and exact duplicates were dropped (none were found).

4.2 Missing values

Placeholder values were converted to real missing values so they are counted honestly. Identifier columns (`Lic No`, `Address`) were labelled `Unknown` because imputing a most-frequent value would invent false information. Capacity and coordinates had no gaps; the median would be used for capacity and the mode for categories if gaps appeared in new data.

4.3 Date parsing

Dates such as `12/21/2073` use the Bikram Sambat calendar, so standard Gregorian parsing would give wrong results. The month, day and year were extracted with a regular expression, and three features were derived:

- `Issue Year (BS)`: the year as recorded.
- `Issue Year (AD approx)`: BS year minus 57 (approximate, because the BS year starts in mid-April).
- `License Term (years)`: validity date minus issue date, with impossible values discarded.

4.4 Outlier treatment

Outliers were detected with the IQR rule (Q1 = 4.2, Q3 = 26.25, IQR = 22.05). Anything above 59.3 MW is an outlier, which gives **74 records (12.9%)**. Nine of the ten largest projects hold only Survey licences.

Decision: the outliers were kept, because they are real large-scale projects rather than entry errors. They are marked with an `Is Capacity Outlier` flag, and a `Log Capacity` column (log of 1 plus capacity) reduces the skew for plots, correlations and models.

4.5 Feature engineering

Feature	Definition
Capacity Category	Small (10 MW or less), Medium (10 to 50), Large (50 to 100), Very Large (above 100)
Province Name	Koshi, Bagmati, Gandaki, Lumbini, Karnali, Sudurpashchim
Is NEA Promoter	1 if the promoter is Nepal Electricity Authority
Is Storage Project	1 if the project name contains "storage"
Issue Year, License Term	Derived from the dates (section 4.3)

4.6 Encoding and scaling

`License Type` and `Province` were one-hot encoded. `Capacity Category` was ordinal-encoded because it has a natural order. Continuous features (log capacity, longitude, latitude, issue year, licence term) were standardized to mean 0 and standard deviation 1. High-cardinality text columns were dropped from the model-ready table.

4.7 Outputs

The pipeline saves `nepal_hydropower_cleaned.csv` (analysis-ready data with all new features), `nepal_hydropower_model_ready.csv` (encoded and scaled) and 12 figures in the `figures/` folder.

5. Exploratory Data Analysis Results

5.1 Capacity distribution

Statistic	Value (MW)
Mean	41.91
Median	9.23
Mode	5.00
Standard deviation	124.86

Statistic	Value (MW)
Minimum / Maximum	0.10 / 1,902.00
Q1 / Q3	4.20 / 26.25

The distribution is strongly right-skewed: a few very large projects pull the mean far above the median. The log transform brings it much closer to symmetric (see `01_outlier_treatment.png`).

5.2 Capacity categories

Category	Records	Share
Small (10 MW or less)	315	55.1%
Medium (10 to 50 MW)	169	29.5%
Large (50 to 100 MW)	42	7.3%
Very Large (above 100 MW)	46	8.0%

5.3 Licence types

Licence type	Records	Share	Mean MW	Median MW	Max MW
Survey	311	54.4%	57.88	9.96	1,902
Generation	180	31.5%	27.66	9.56	456
Operation	81	14.2%	12.26	4.50	144

Survey projects have the highest average, yet their median is close to that of Generation projects. The higher average is caused by a small number of very large proposals. Operation projects are the smallest.

5.3.1 Geography

Province	Records
Province 4 (Gandaki)	182
Province 3 (Bagmati)	157

Province	Records
Province 1 (Koshi)	155
Province 7 (Sudurpashchim)	28
Province 6 (Karnali)	27
Province 5 (Lumbini)	23

Provinces 4, 3 and 1 hold 494 of 572 records (86%). The busiest districts are Dolakha and Sindhupalchok (44 records each) and Kaski (43), followed by Taplejung (38), Sankhuwasabha (35) and Lamjung (34). The geographic scatter plot (`08_geographic_distribution.png`) shows projects concentrated along the Himalayan belt.

5.4 Promoters and rivers

Development is fragmented: 511 distinct promoters appear in 572 records. Nepal Electricity Authority leads with 28 records, and the next most frequent promoter has only 4. Among the ten largest projects, four are on the Karnali river, two on the Arun and two on the Tama Koshi.

5.5 Correlations

The original correlation matrix of capacity, longitude and latitude gave these values.

Pair	Correlation
Capacity and latitude	0.18
Capacity and longitude	-0.13
Longitude and latitude	-0.92

The strong negative link between longitude and latitude only reflects Nepal's south-east to north-west shape. Capacity has weak correlation with location, so **location alone does not explain project size**. The notebook also reports Spearman correlations, which suit skewed data better than Pearson.

6. Feature Selection

Features were ranked against `Log Capacity` using two methods: mutual information (any dependency, including non-linear) and random-forest importance (interactions). Features derived from capacity (`Capacity (MW)`, the ordinal category and the outlier flag) were excluded to avoid target leakage. The top six features by average rank are selected, and a five-fold cross-validated R-squared shows how much size can be explained by these features. The ranking and the R-squared value are produced when the notebook is run (`12_feature_importance.png`). Identifier columns such as `Lic No`, `Address` and `Project` were excluded because they carry no generalizable signal.

7. Key Findings

1. **Size:** most projects are small. The median is 9.2 MW, and 55% of records are 10 MW or below. Averages are misleading because a few giant proposals dominate them.
2. **Pipeline:** 54% of records are Survey licences and only 14% are Operation licences, so most of the register is still at an early stage.
3. **Proposals vs built capacity:** the largest figures (such as Mugu Karnali Storage at 1,902 MW) are survey-stage proposals, not operating plants.
4. **Geography:** three provinces hold 86% of all records, while Provinces 5, 6 and 7 together hold only 78.
5. **Structure:** the market is fragmented among hundreds of small promoters, with the state utility as the single largest.
6. **Data quality:** placeholder values, BS-calendar dates and inconsistent river naming make the raw file unreliable without cleaning.

8. Recommendations

1. Report medians and capacity categories instead of means alone.
2. Track the conversion from Survey to Generation to Operation licences over time.
3. Examine why Provinces 5, 6 and 7 are under-represented, and why Province 2 is missing.
4. Improve data capture: use true blanks instead of `0`, store dates in a single calendar, and standardize river names.
5. Add physical variables (river discharge, head, catchment area) to build a capacity model that works.

9. Limitations

- The data is a licence register snapshot with no generation output or commissioning dates.
- Survey-stage capacity is a proposal that may never be built, so totals mix proposed and licensed capacity.
- The AD year is approximate (BS minus 57), and 40 validity values were unusable.
- Some river names are inconsistent in the source, which may split totals for the same river.

10. Reproducibility

Install the packages in `requirements.txt`, place `hydropower.csv` in the `data/` folder and run `notebooks/Nepal_Hydropower_Data_Analysis.ipynb` from top to bottom. All outputs are regenerated in `data/` and `figures/`. Setup and GitHub instructions are in `README.md`.